

Group by



Group by

Klauzula instrukcji SELECT, która dzieli wynik zapytania na grupy wierszy, zwykle poprzez wykonanie jednej lub więcej agregacji na każdej grupie. Instrukcja SELECT zwraca jeden wiersz na grupę.

Dozwolone kombinacje:

- `SELECT ColumnA, ColumnB FROM T GROUP BY ColumnA, ColumnB;`
- `SELECT ColumnA + ColumnB FROM T GROUP BY ColumnA, ColumnB;`
- `SELECT ColumnA + ColumnB FROM T GROUP BY ColumnA + ColumnB;`
- `SELECT ColumnA + ColumnB + constant FROM T GROUP BY ColumnA, ColumnB;`

Niedozwolone kombinacje:

- `SELECT ColumnA, ColumnB FROM T GROUP BY ColumnA + ColumnB;`
- `SELECT ColumnA + constant + ColumnB FROM T GROUP BY ColumnA + ColumnB;`

Grouping function - Count

```
select  
    count(*)  ilosc  
from employees
```

	ilosc
1	107

Count
(column with
null)

	ilosc
1	35

```
select
    count(commission_pct) ilosc
from employees
```

Count with distinct

Results		Messages
	dept	dist_dept
1	106	11

```
select
    count(department_id) dept,
    count(distinct department_id) dist_dept
from employees
```



Avg

	avg	
1	6461.682242	

```
select  
  avg(salary) avg  
from employees
```

Avg (column with null)

	avg	
1	4	

```
select avg(t1.a) avg
from(
  select
    null as a
  union
  select
    4 as a) t1
```



	max	
1	24000.00	

```
select  
  max(salary) max  
from employees
```




Min

	min	
1	2100.00	

```
select
  min(salary) min
from employees
```

Min(column
with null)

	min	
1	0.10	

```
select  
    min(commission_pct) min  
from employees
```



	suma	
1	691400.00	

```
select
  sum(salary) suma
from employees
```

Group by

	department_id	suma
1	NULL	7000.00
2	10	4400.00
3	20	19000.00
4	30	24900.00
5	40	6500.00
6	50	156400.00
7	60	28800.00
8	70	10000.00
9	80	304500.00
10	90	58000.00
11	100	51600.00
12	110	20300.00

```
select
    department_id,
    Sum(salary) suma
from employees
group by department_id
```

Group by
(specific
error)

Messages

```
Msg 8120, Level 16, State 1, Line 3  
Column 'employees.job_id' is invalid in the select list  
because it is not contained in either an aggregate function or the GROUP BY clause.
```

```
select  
    department_id,  
    job_id,  
    Sum(salary) suma  
from employees  
group by department_id
```

Group by
(many
groups)

	department_id	job_id	suma
1	110	AC_ACCOUNT	8300.00
2	110	AC_MGR	12000.00
3	10	AD_ASST	4400.00
4	90	AD PRES	24000.00
5	90	AD_VP	34000.00
6	100	FI_ACCOUNT	39600.00
7	100	FI_MGR	12000.00
8	40	HR_REP	6500.00
9	60	IT_PROG	28800.00

```
select
    department_id,
    job_id,
    Sum(salary) suma
from employees
group by department_id, job_id
```

Group by with having

	department_id	suma
1	20	19000.00
2	30	24900.00
3	50	156400.00
4	60	28800.00
5	80	304500.00
6	90	58000.00
7	100	51600.00
8	110	20300.00

```
select
    department_id,
    Sum(salary) suma
from employees
group by department_id
having Sum(salary) > 10000
```

Group by
(specific
error)

Messages

Msg 147, Level 15, State 1, Line 5
An aggregate may not appear in the WHERE clause
unless it is in a subquery contained in a HAVING clause or a select list,
and the column being aggregated is an outer reference.

```
select
    department_id,
    Sum(salary) suma
from employees
where Sum(salary) > 10000
group by department_id
```


Group by with where and having

	department_id	suma
1	20	19000.00
2	30	24900.00
3	50	156400.00

```
select
    department_id,
    Sum(salary) suma
from employees
where department_id between 10 and 50
group by department_id
having Sum(salary) > 10000
```